Project Report

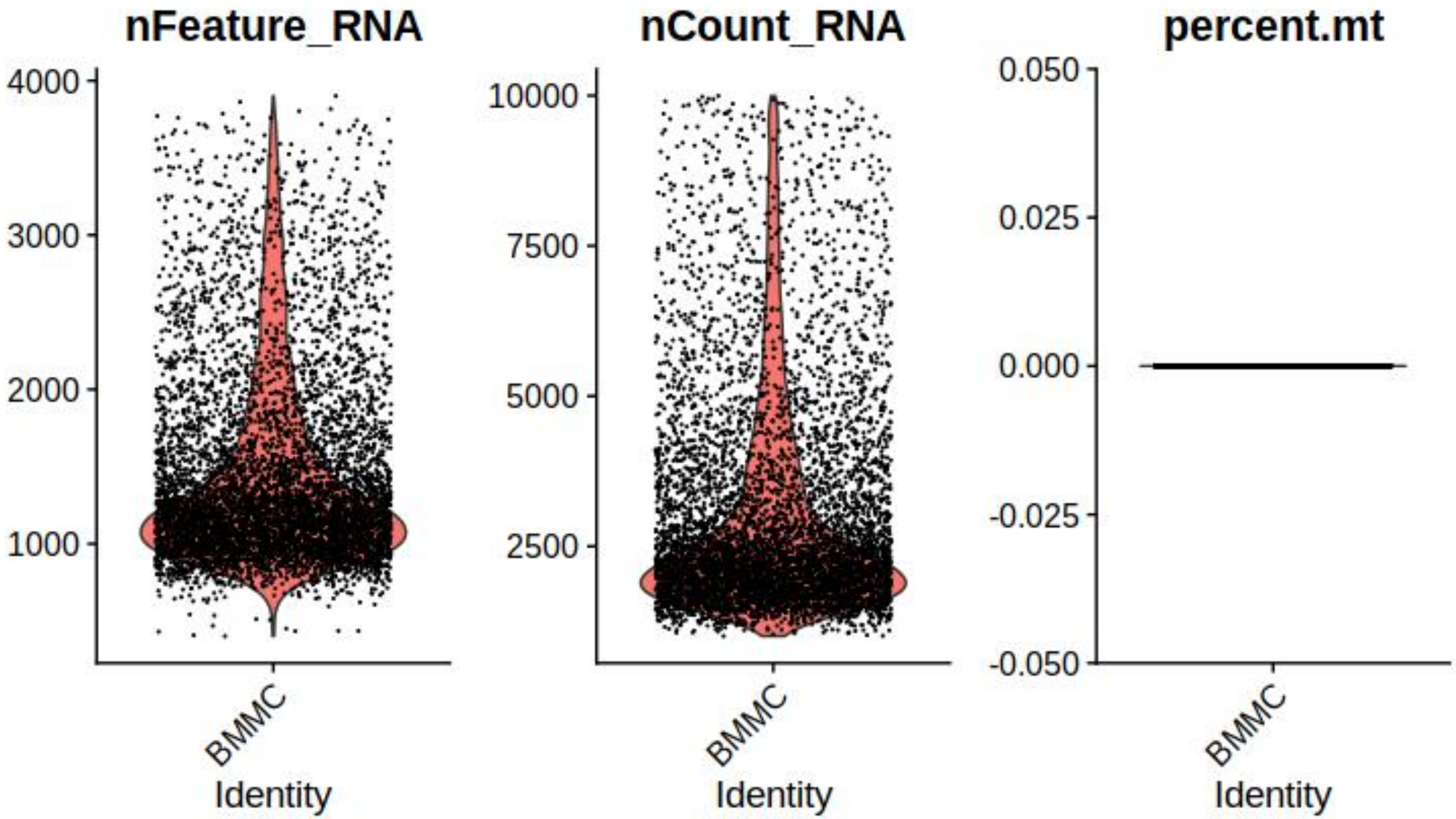
How many cells are in each sample

Sample	Number of cells
BMMC-D1T1	6270
BMMC-D1T2	6332
CD34-D2T1	2424
CD34-D3T1	5752

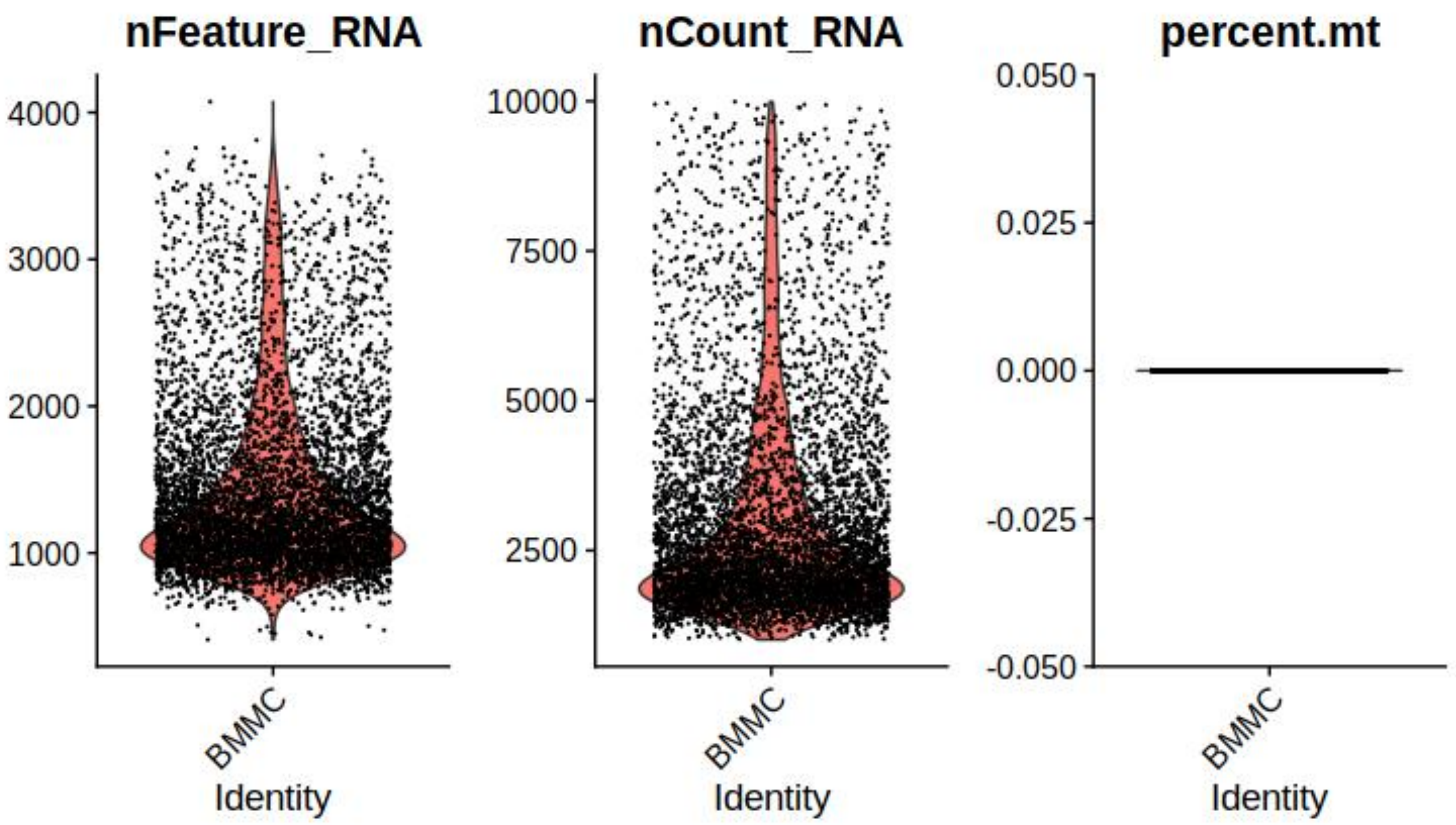
How many genes are in the expression matrices

Sample	Number of Genes
BMMC-D1T1	14065
BMMC-D1T2	14084
CD34-D2T1	13441
CD34-D3T1	13432

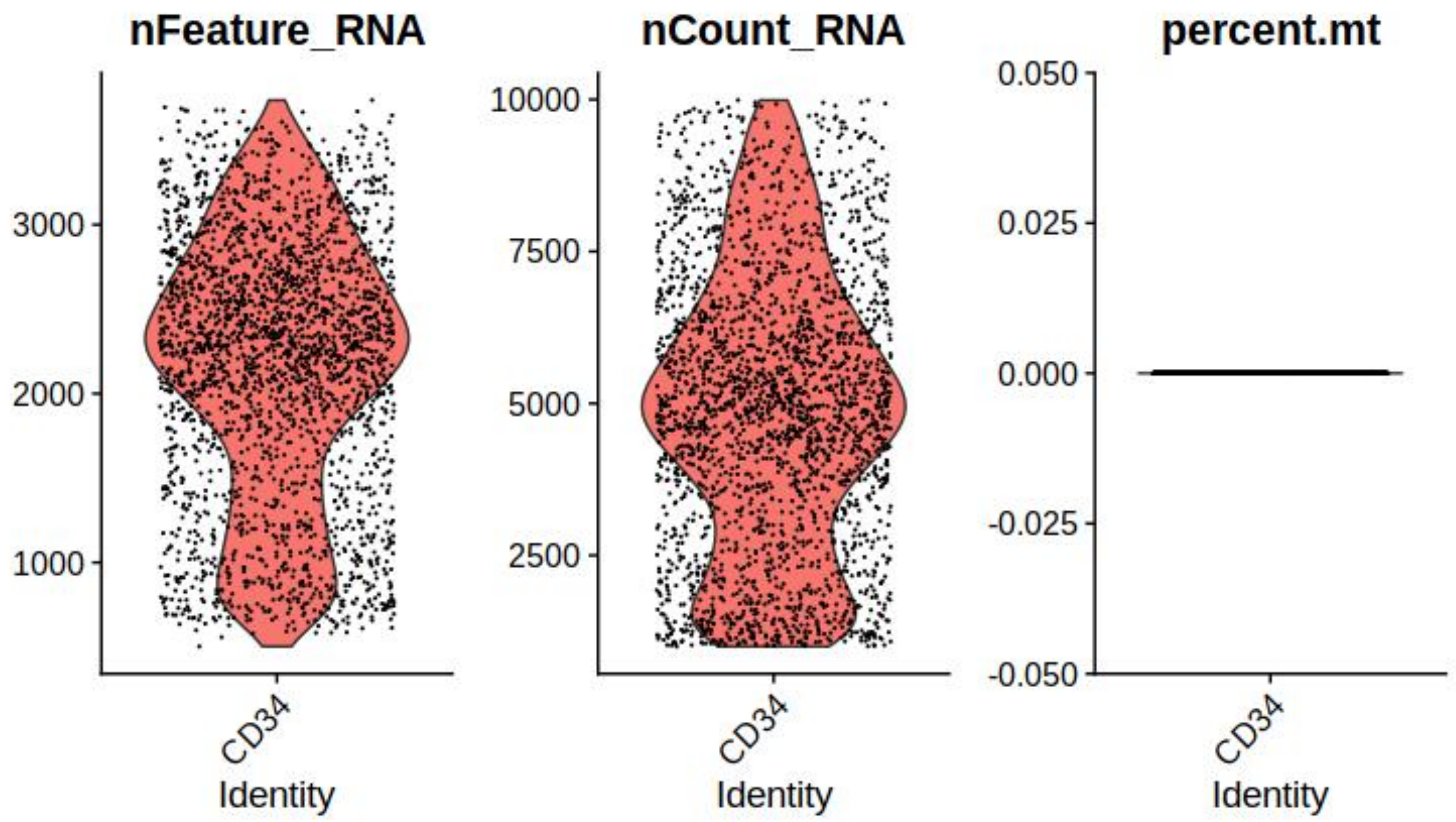
VlnPlot of BMMC_D1T1



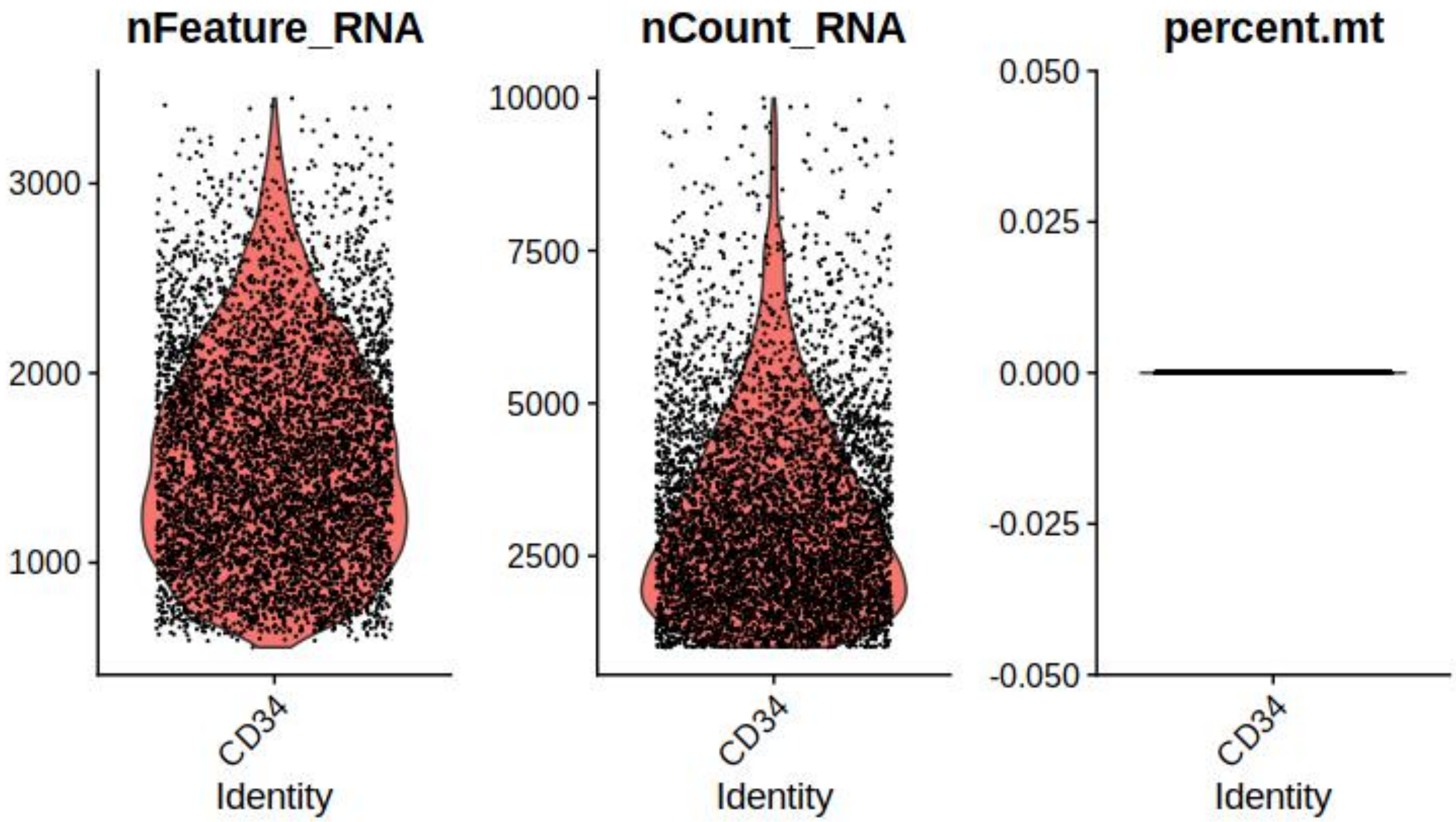
VlnPlot of BMMC_D1T2



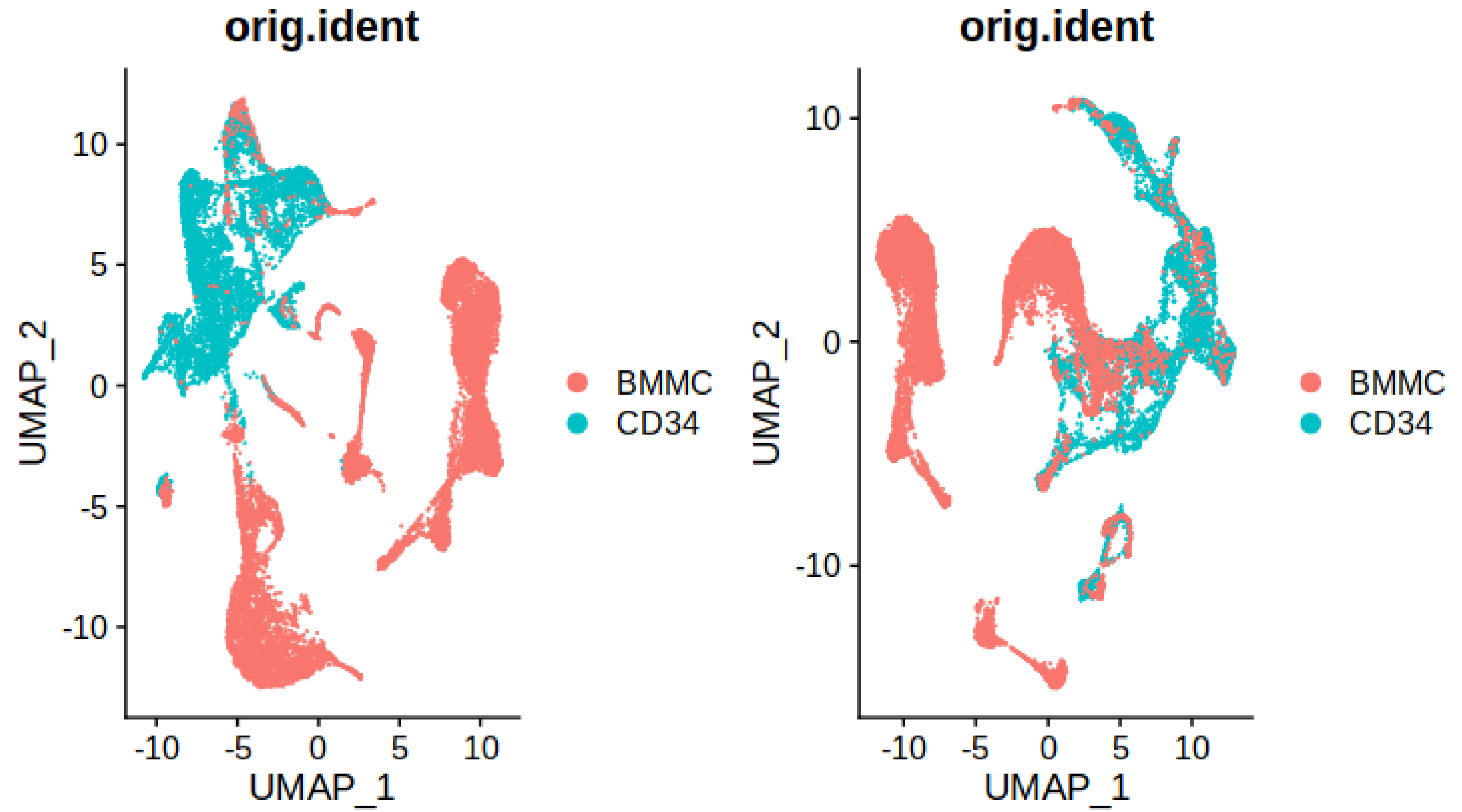
VlnPlot of CD34_D2T1



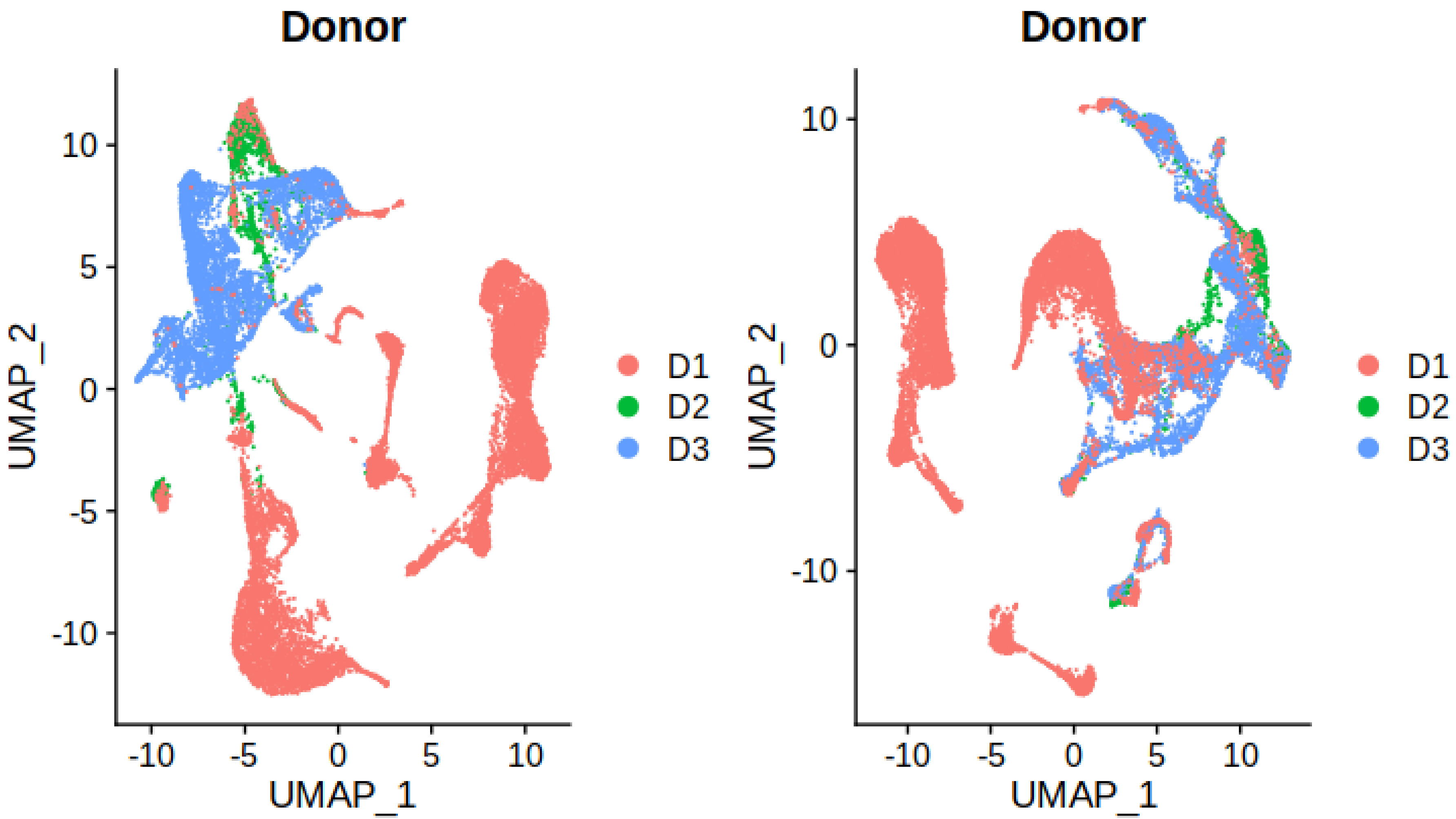
VlnPlot of CD34_D3T1



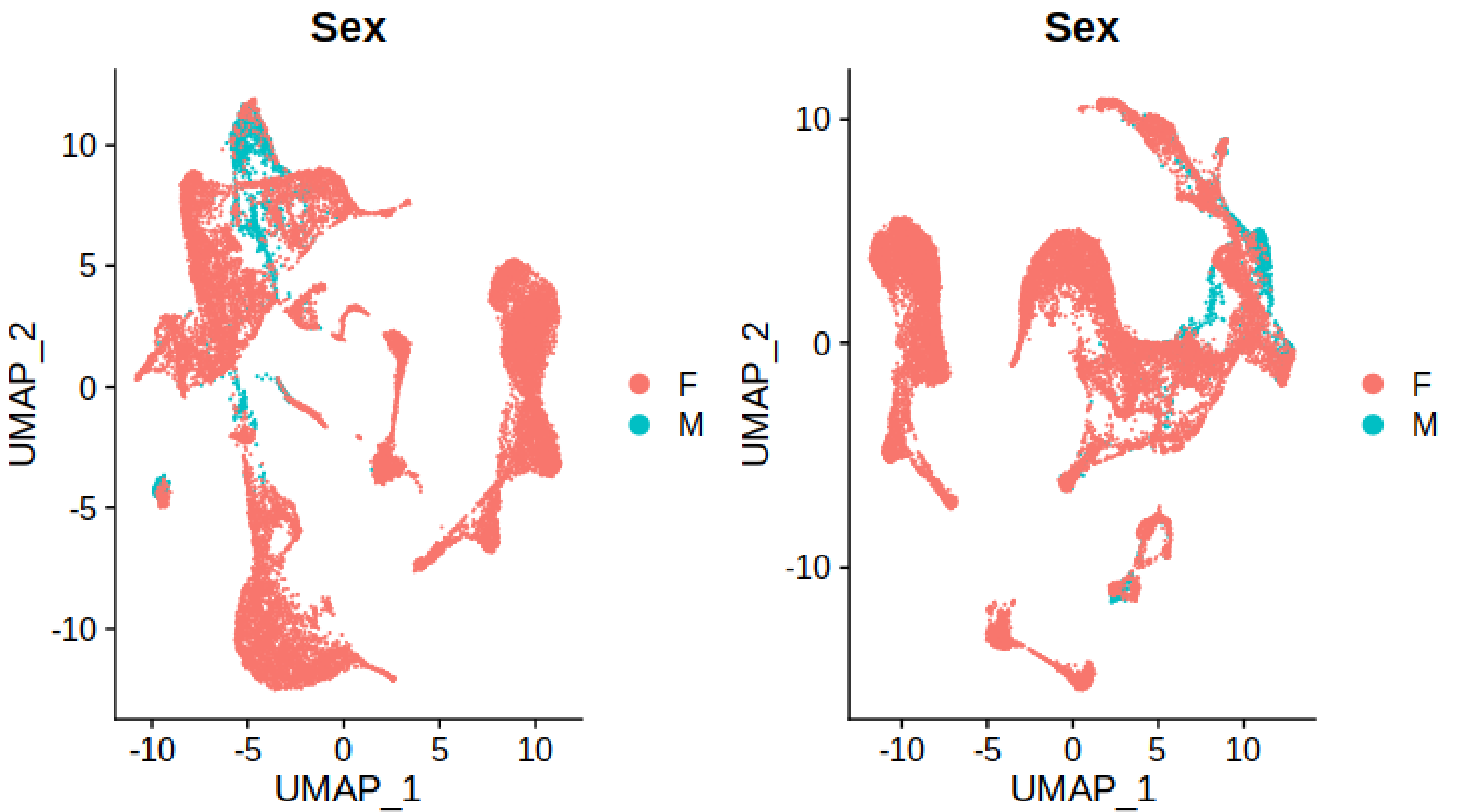
Batch-Correction Plot orig.ident



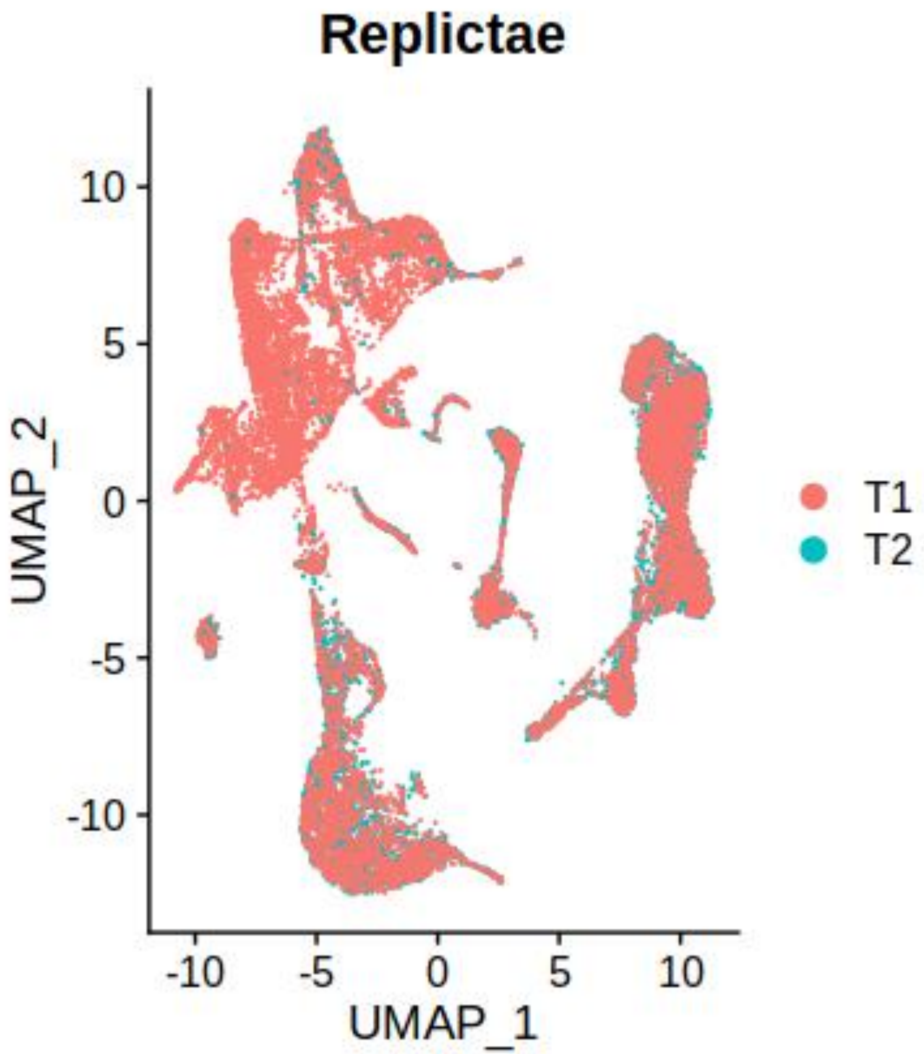
Batch-Correction Plot Donor

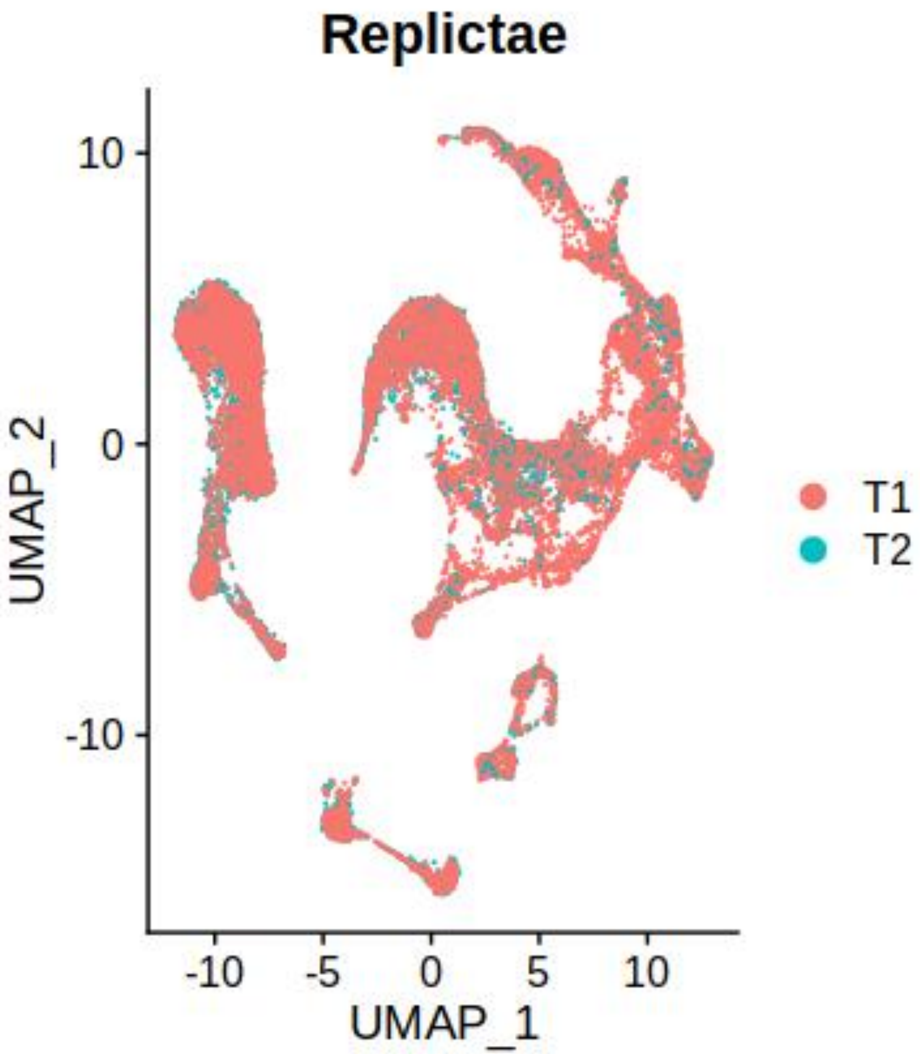


Batch-Correction Plot Sex

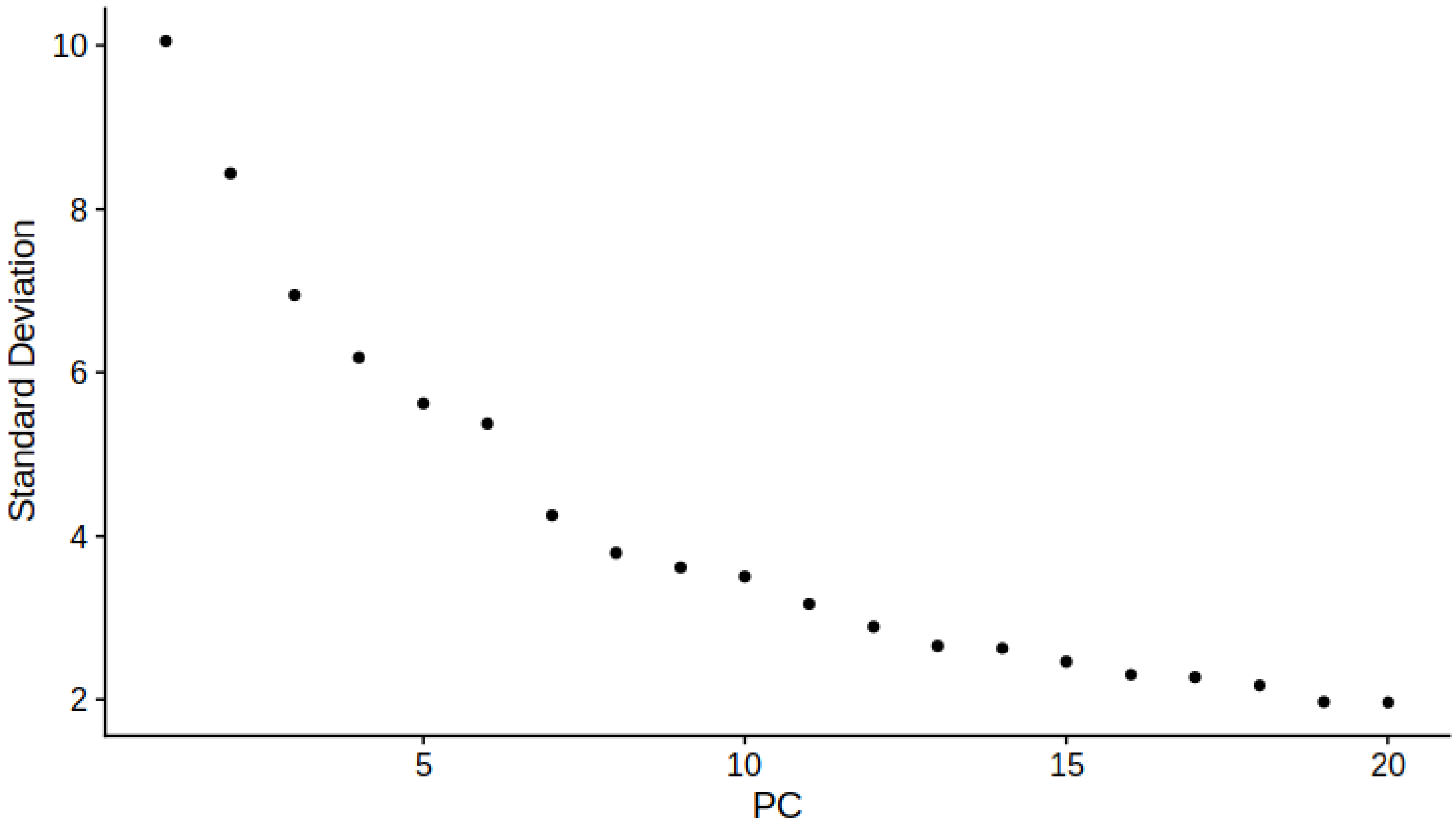


Batch-Correction Plot Replicate



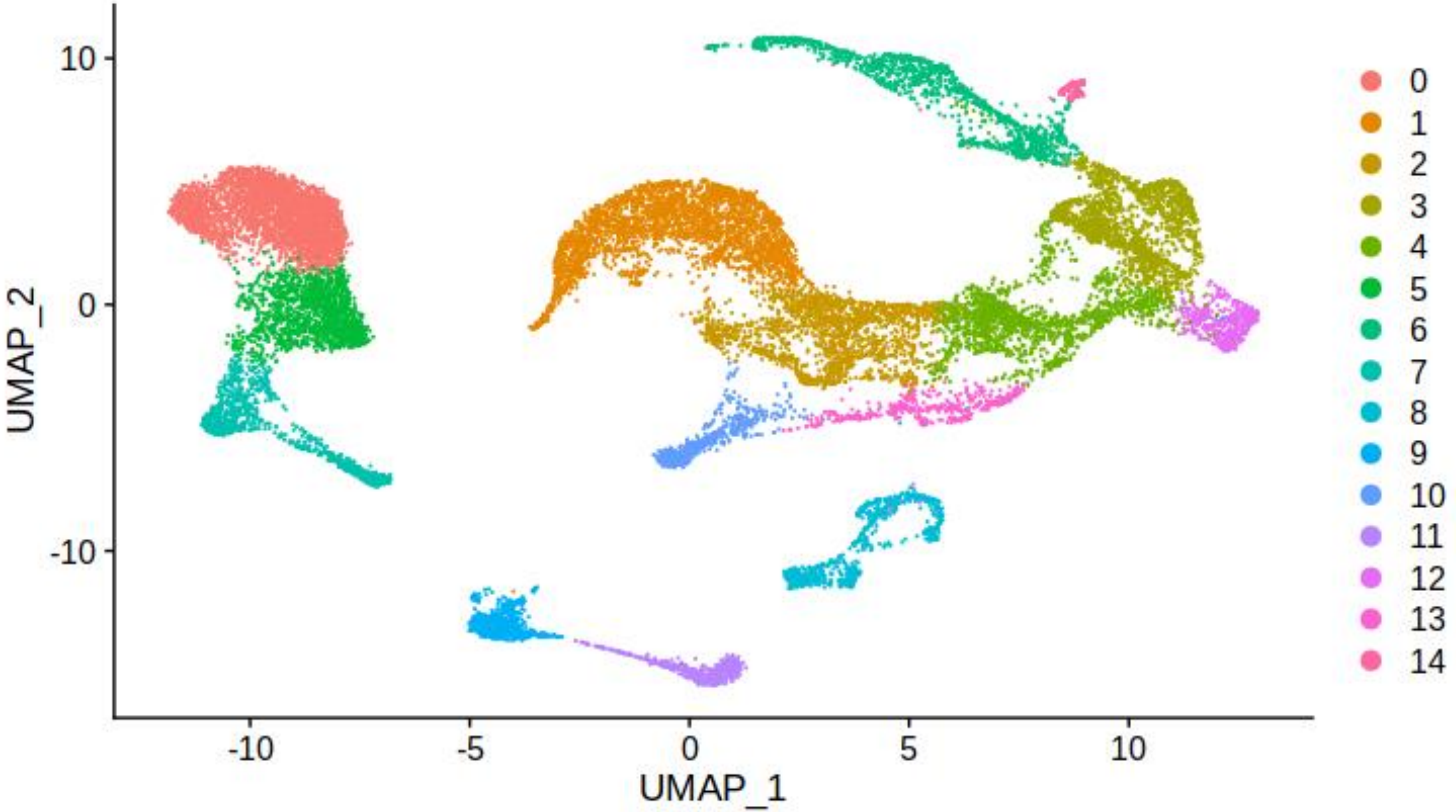


Dimensionality Reduction : PCA Plot

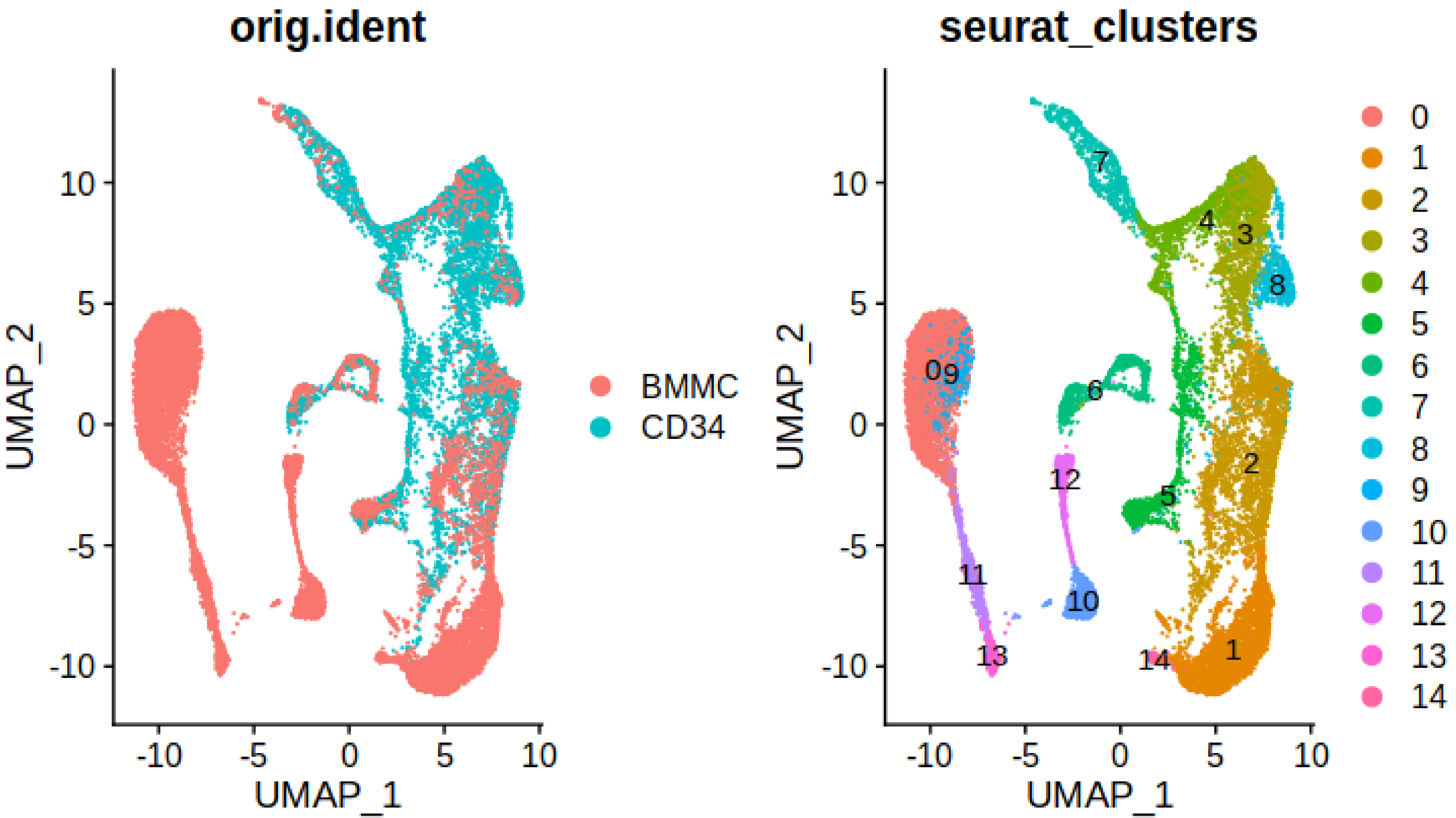


The "elbow" in the plot (around 6–7 PCs, as mentioned) indicates where the explained variance begins to level off. This suggests that the majority of the signal in the data is captured within the first 10 PCs, making it a suitable choice for further analysis.

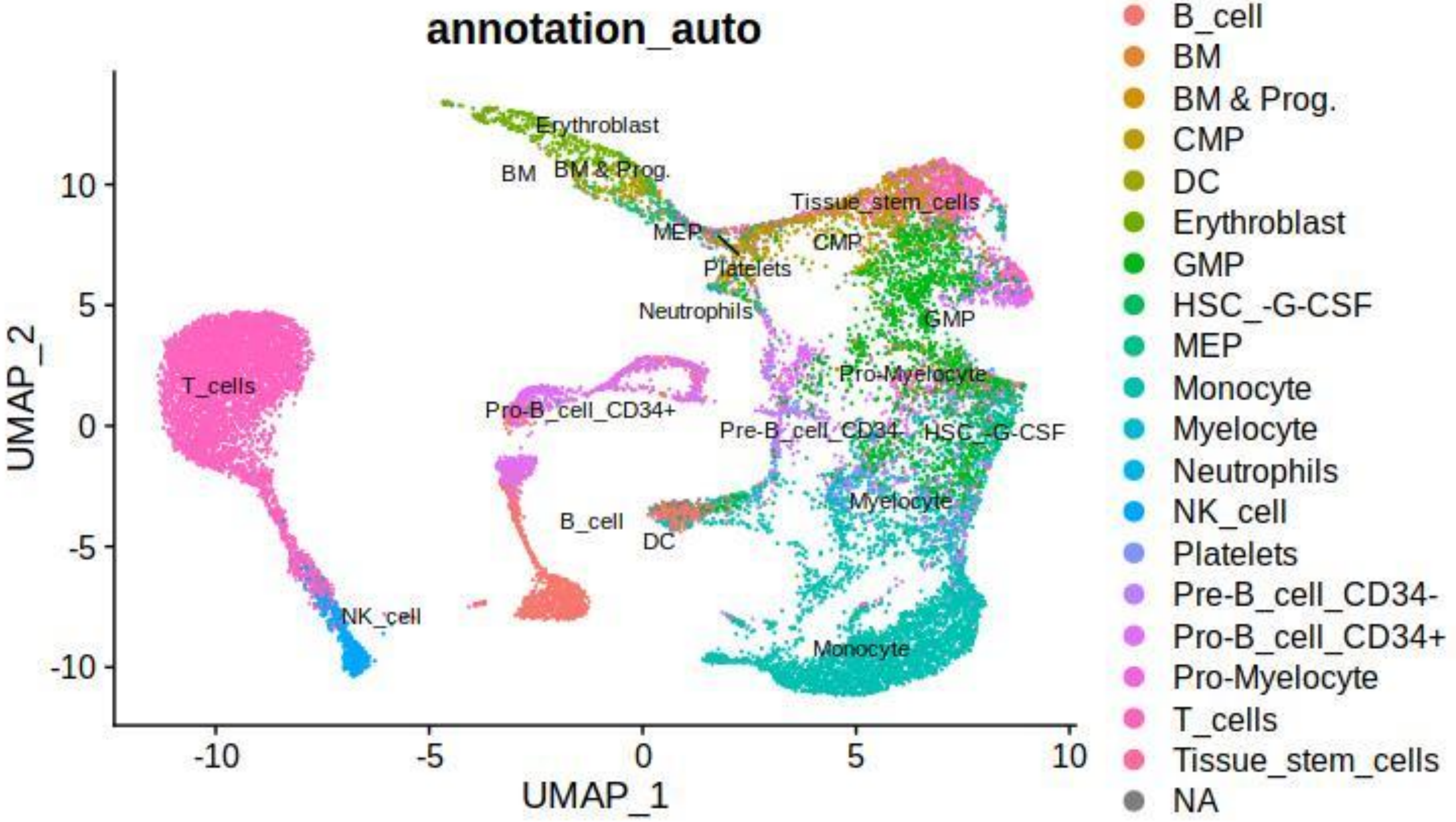
Dimensionality Reduction: UMAP



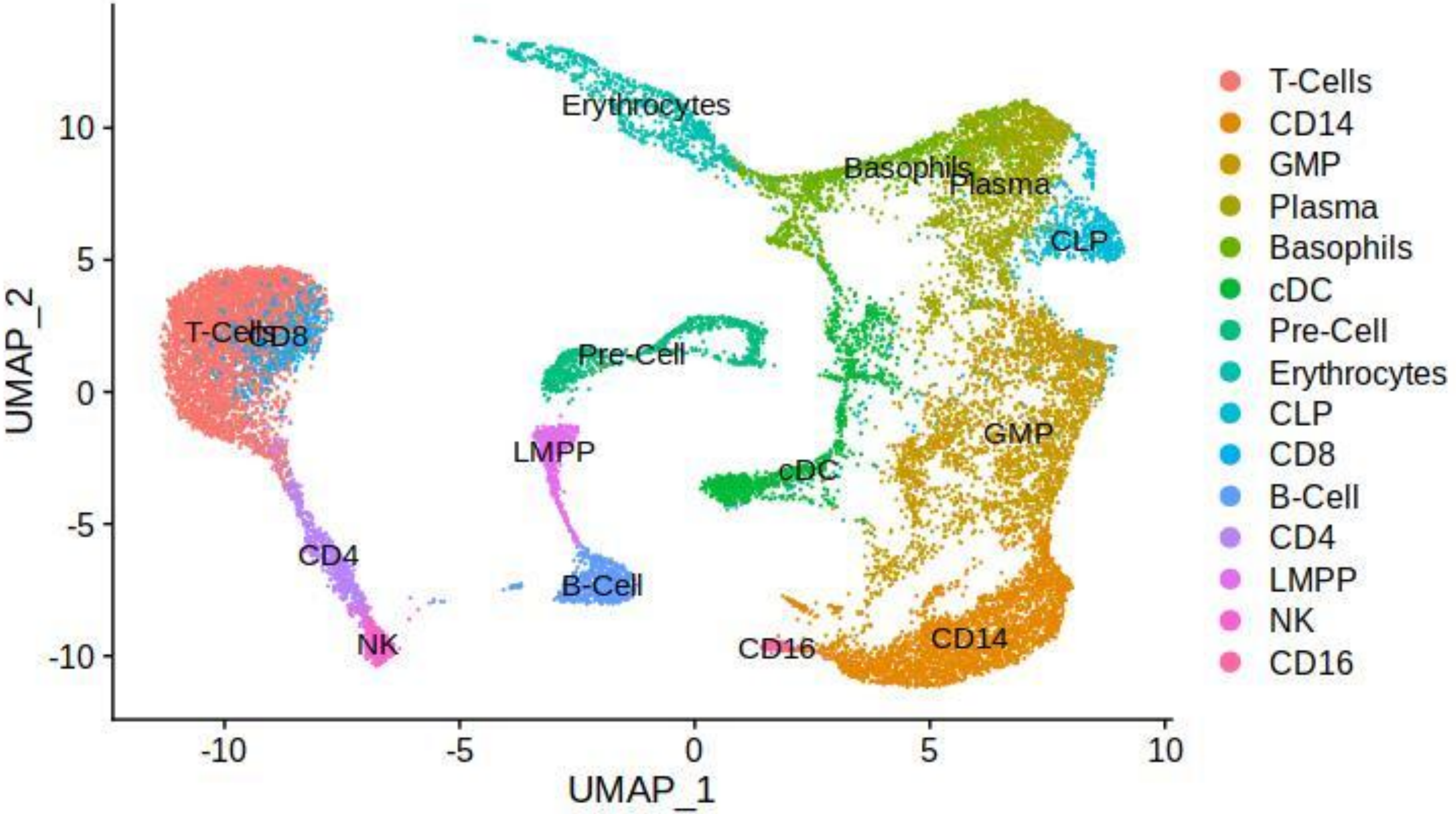
Clustering


orig.ident
seurat_clusters
UMAP_2
UMAP_1
10
5
0
-5
-10
BMMC
CD34
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14

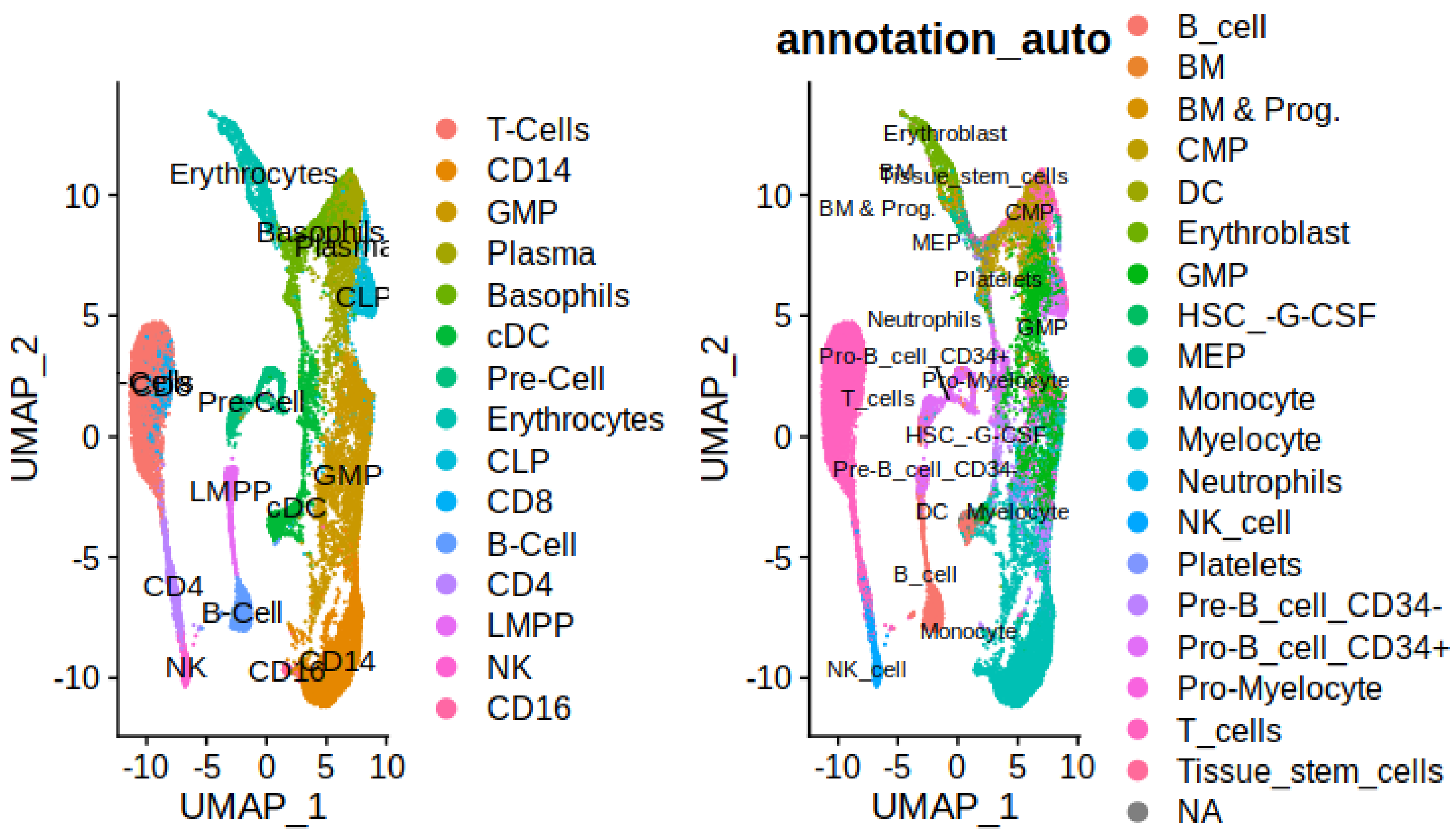
Auto Annotation



Manual Annotation

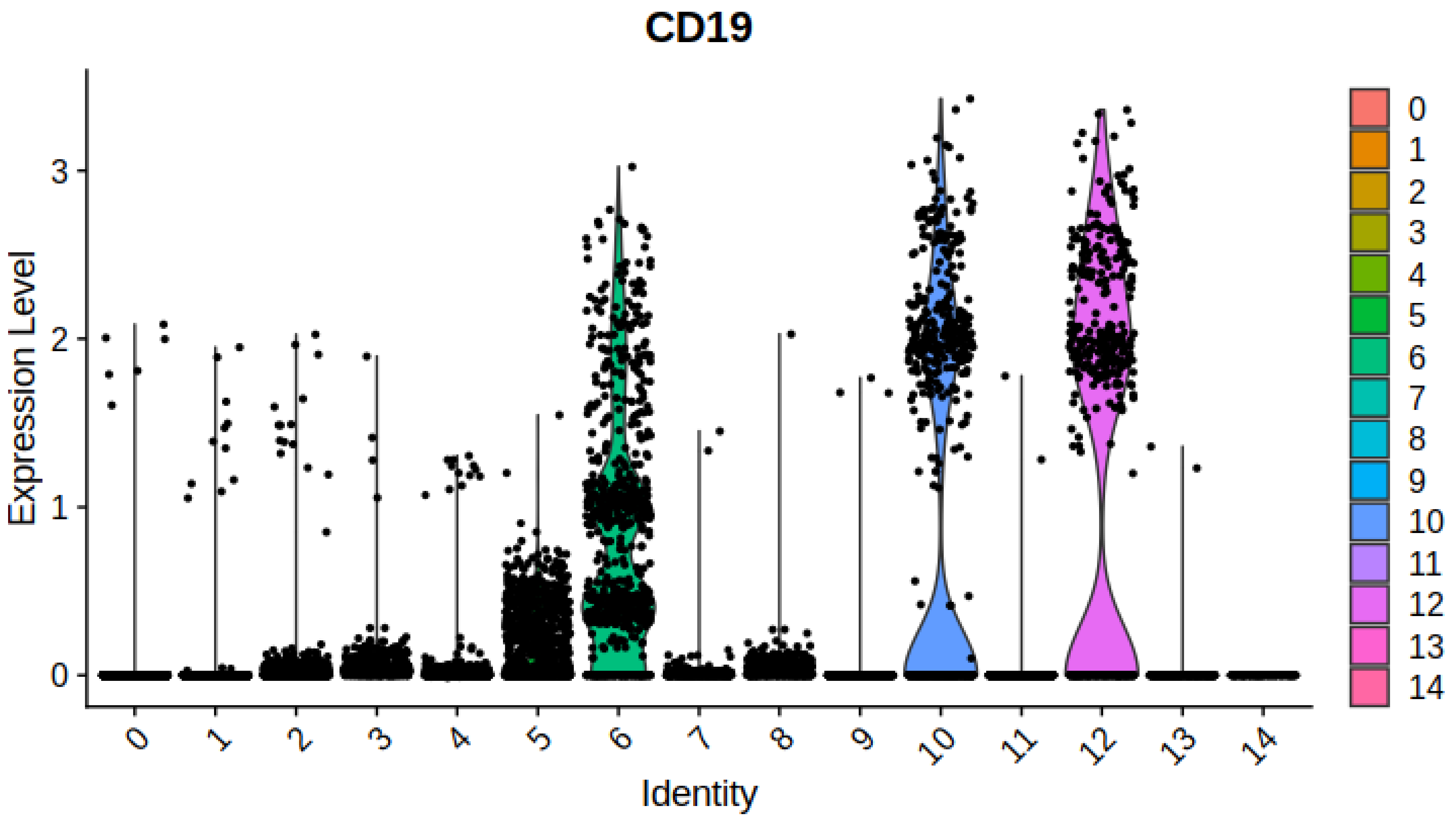


Comparison Manual vs Automatic

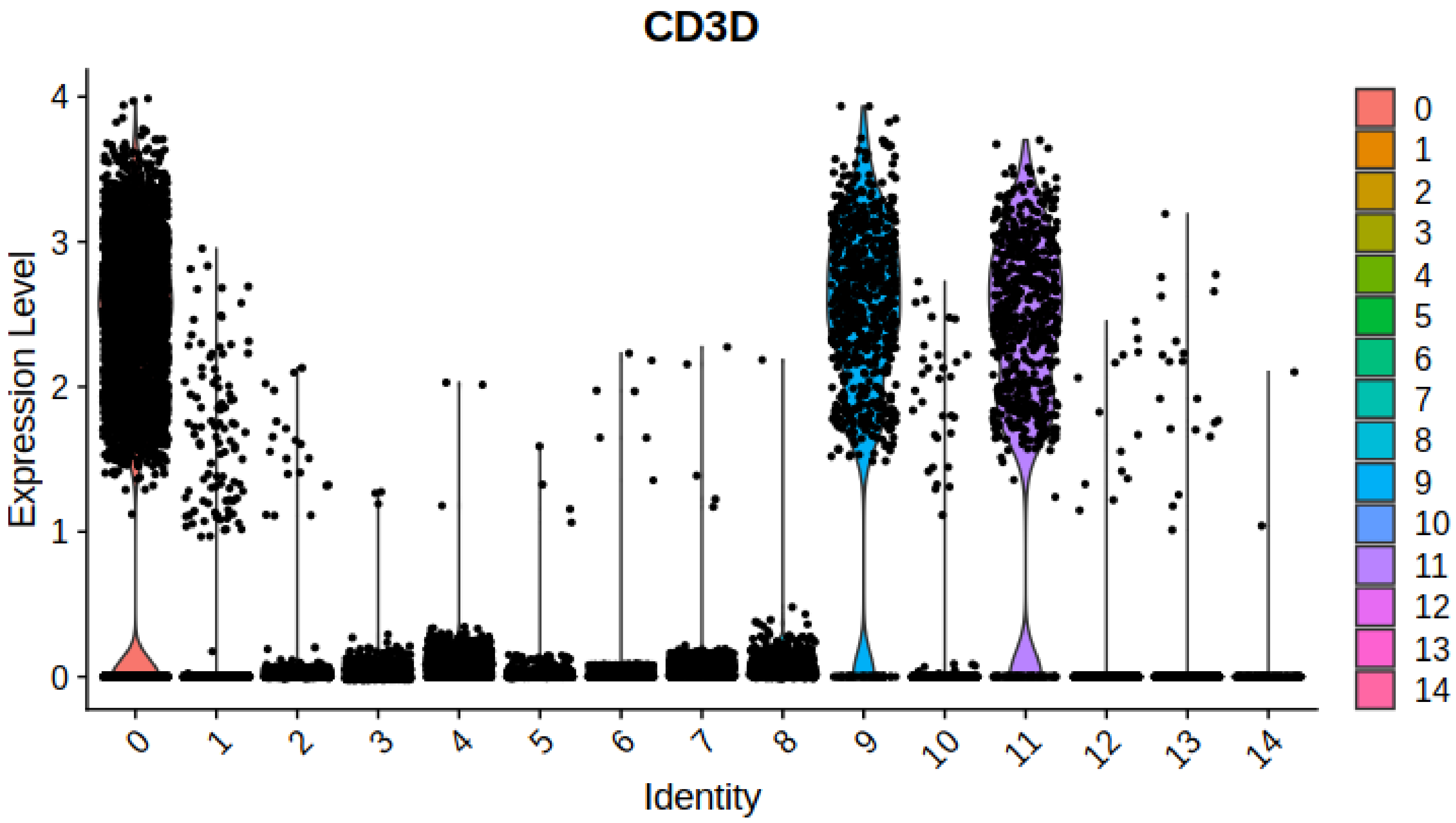


There was similarity between the distribution of the cell types over the clusters. However, In automatic annotation we can see that there more cell types lablelled on the clusters like for example T-cell cluster is one cluster in automatic annotation while in the manual annotation it is treated as three clusters. Which make the number of annotated cells in the automatic more than the manual one.

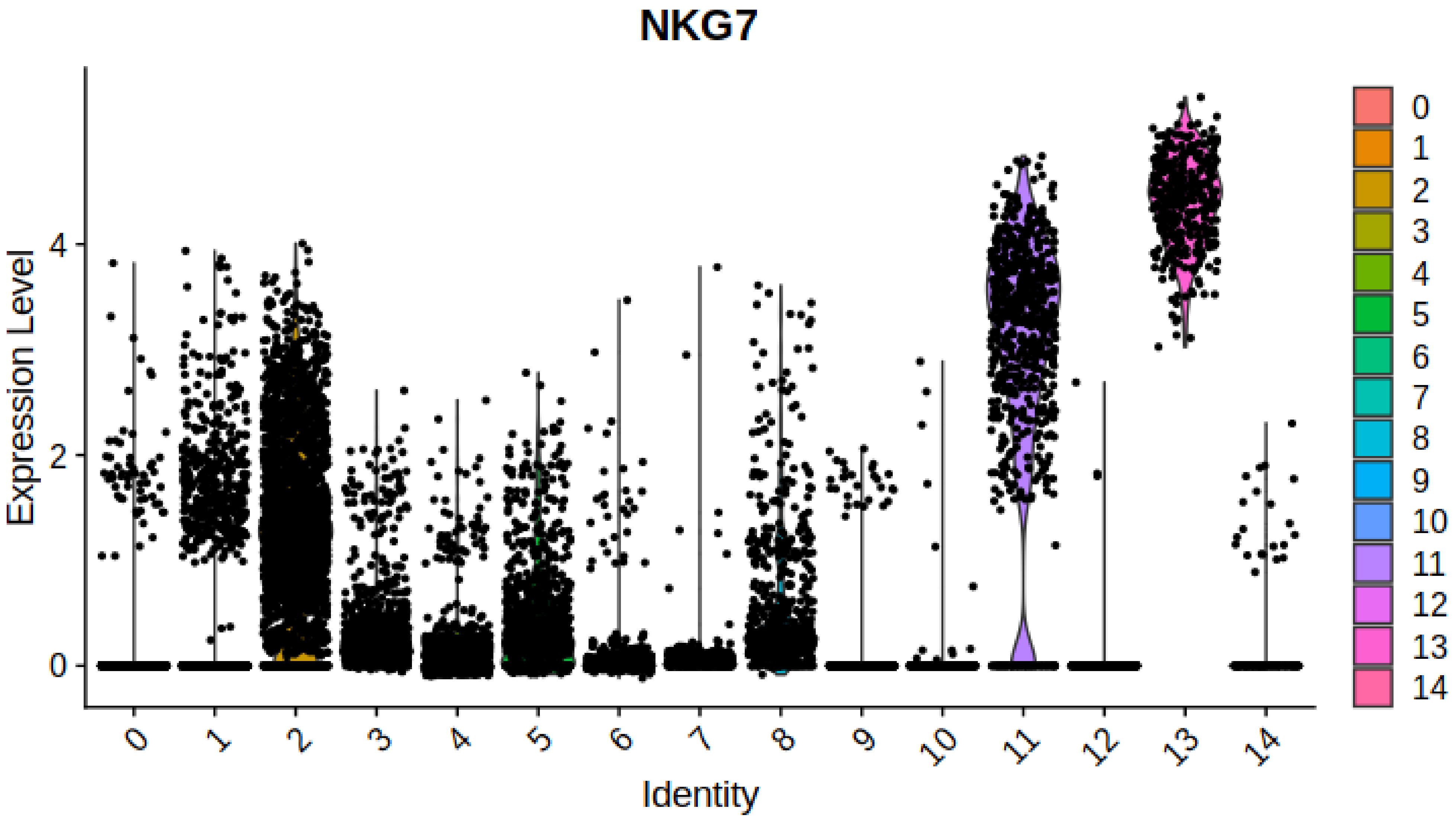
Differential Expression Analysis



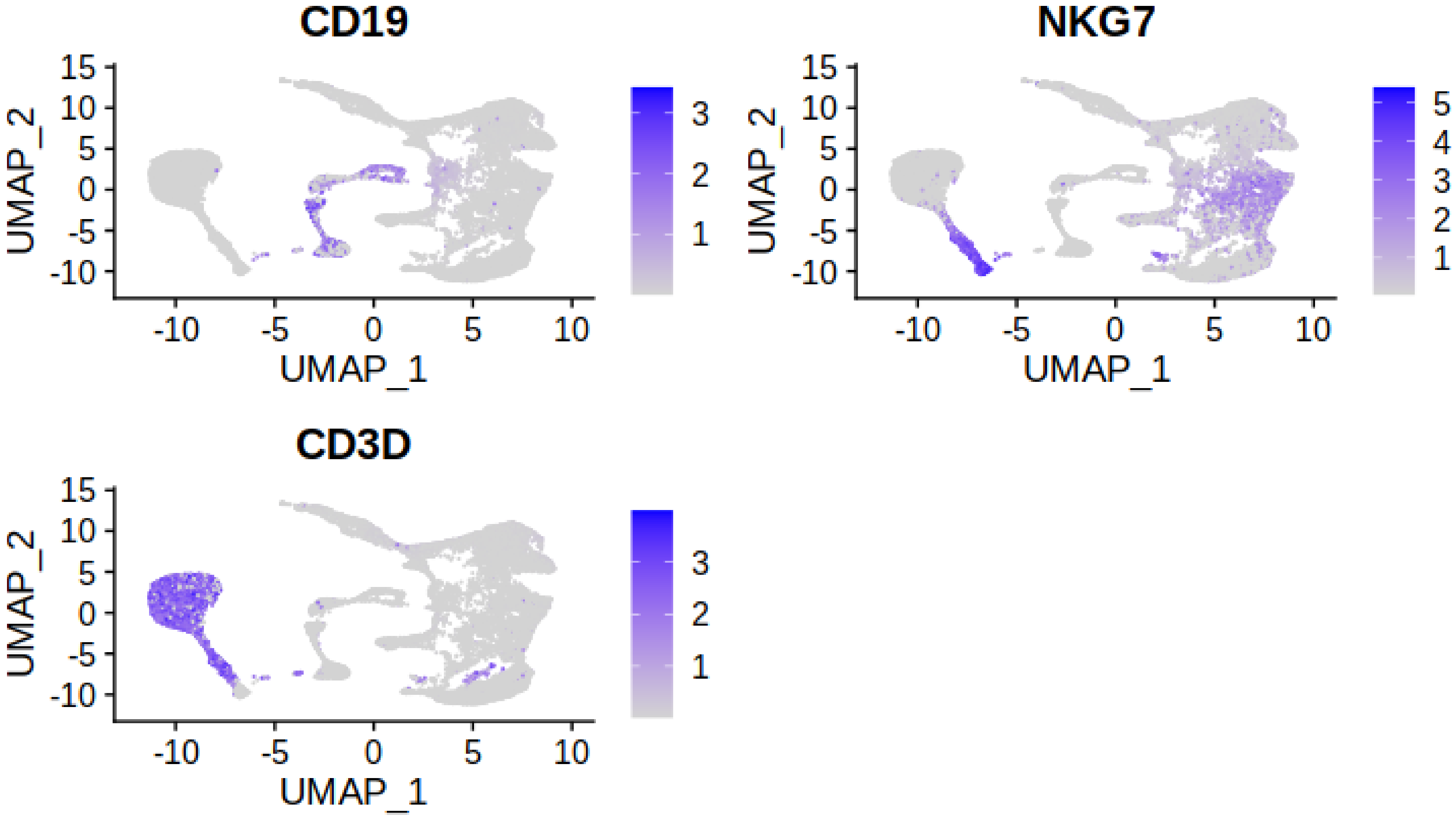
Differential Expression Analysis



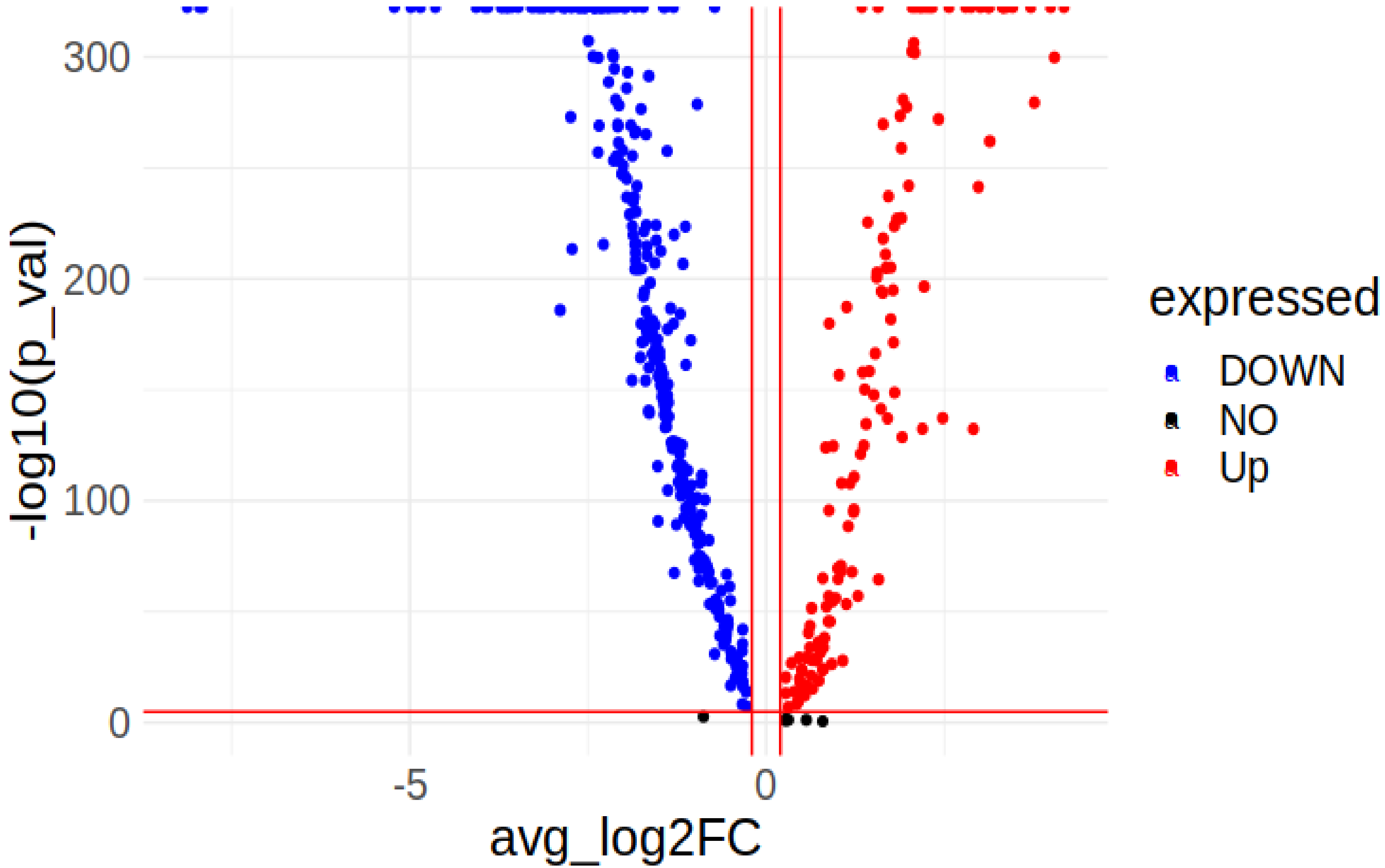
Differential Expression Analysis



Differential Expression Analysis

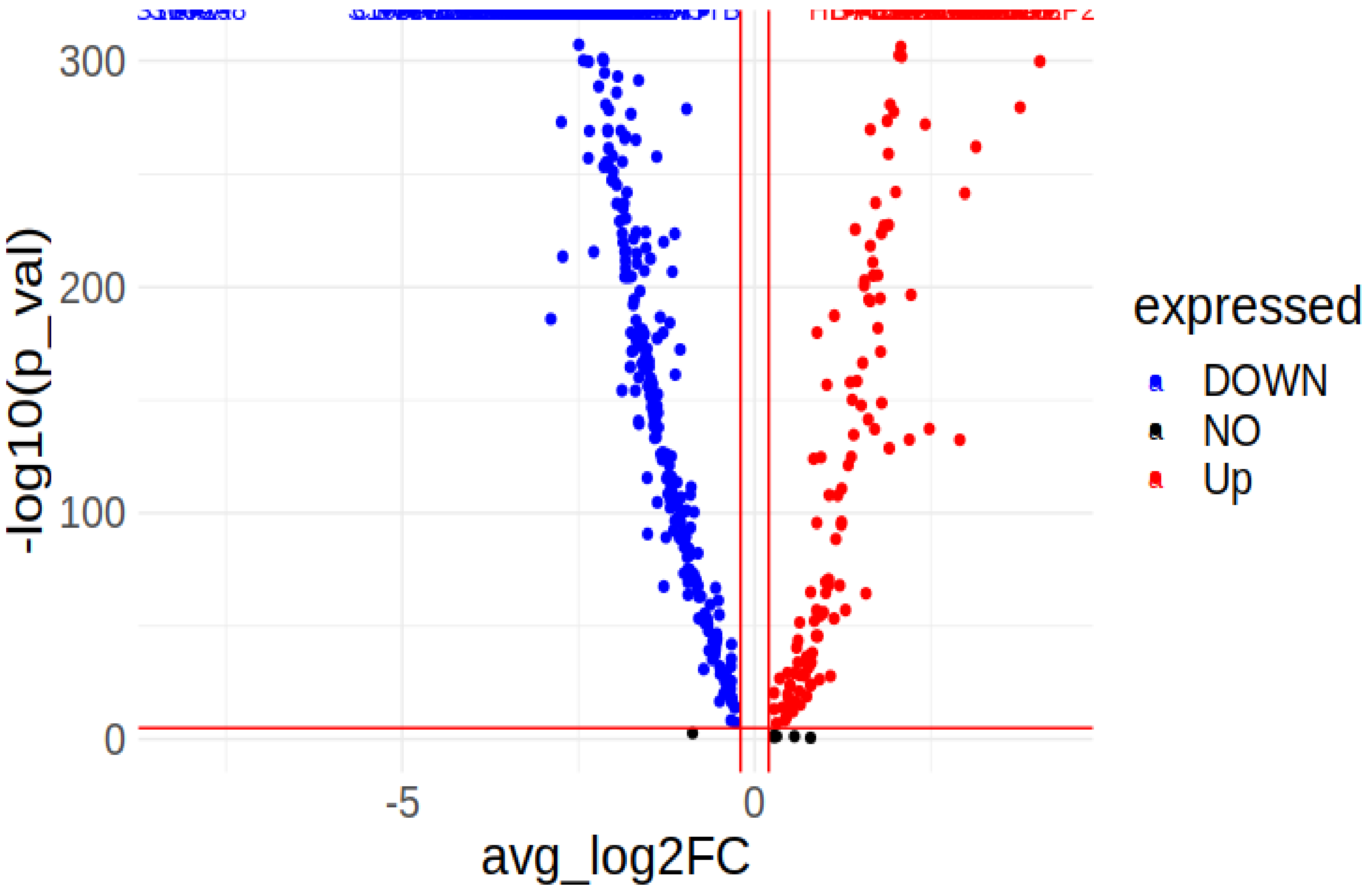


Differential Expression Analysis



B-Cells Vs T-Cells

Differential Expression Analysis



<u>CD4 T Cells Vs CD14 Monocytes</u>